A320 TYPE RATING · SIM TRAINING

Trainee A + Trainee B syllabus

18 sessions | MFTD/FBS 2–9 | FFS 1–10

Overview | MFTD/FBS 2 | MFTD/FBS 3 | MFTD/FBS 4 | MFTD/FBS 5 | MFTD/FBS 6 | MFTD/FBS 7 | MFTD/FBS 8 | MFTD/FBS 9 | FFS 1 | FFS 2 | FFS 3 | FFS 4 | FFS 5 | FFS 6 | FFS 7 | FFS 8 | FFS 9 | FFS 10

Course overview

Trainee A emphasis

Managed climb and FMGC work, APU MEL and crossbleed starts, fuel-feed and IDG faults, ADR/IR faults, dual-engine failure, overweight approaches, and severe engine damage.

Trainee B emphasis

Ground-speed mini and braking, start-valve and smoke events, FMGC resets, sidestick priority, emergency electrical configuration, contaminated-runway and gear work, and gravity gear extension.

Common ground

Both cover takeoffs, approaches, landings, stalls, UPRT and one-engine-inoperative handling. FFS 8 has the same UPRT exercise sequence; FFS 10 has the same skill-test exercise sequence, with different weather.

MFTD/FBS 2

Packs OFF and managed-climb/FMGC work for A; packs ON for B.

MFTD/FBS 3

APU MEL and crossbleed start for A; ground-speed mini and braking for B.

MFTD/FBS 4

Fuel-feed/IDG/relight work for A; FMGC/FPV/cabin-pressure work for B.

MFTD/FBS 5

Hot start, cargo smoke and evacuation for A; engine/hydraulic/braking failures for B.

MFTD/FBS 6

Unreliable speed, ADR and fuel leak for A; start-valve and smoke for B.

MFTD/FBS 7

A adds relight/RNP and V2 takeoff failure; B adds fire and evacuation.

MFTD/FBS 8

A focuses on OEI approaches and RTO; B on FMGC resets and emergency electrical.

MFTD/FBS 9

Routes run in opposite directions; A focuses on engine fire/OEI, B on contamination and gear.

FFS 1

A introduces RMP/ACP; B adds sidestick priority. Handling exercises overlap.

FFS 2

A adds 3D RNP; both cover stalls, visual circuits and 2D RNP.

FFS 3

A adds cold-weather preflight/taxi; arrival handling largely overlaps.

FFS 4

IR faults and suspected damage for A; ADR faults and no-damage descent for B.

FFS 5

Dual-engine failure/RNP for A; unreliable speed/ manual ILS for B. Alternate branch depends on crew roles.

FFS 6

A adds G+B hydraulics, flaps fault and RNP; B adds slats fault and gravity gear.

FFS 7

Opposite routes; A focuses on FCU/overweight/ severe damage, B on windshear/fuel leak/APU fire.

FFS 8

The UPRT exercise sequence is the same for both trainees.

FFS 9

Opposite routes; EDTO failures differ at EEP, ETOPS entry and ETP.

FFS 10

The skill-test exercise sequence matches; departure and arrival weather differ.

Key A only B only Both Conditional Amber rows mark different values or topics.

SESSION 1 OF 18

MFTD/FBS 2

A VTCC / VTBS | B VTCC / VTBS

Combined practice 7 items

TAKEOFF & DEPARTURE

- Take-off - instrument, packs OFF — A
- Take-off – instrument, packs ON — B

APPROACHES & GO-AROUNDS

- Approach - ILS, with AP — Both

LANDING

- Landing - normal, autoland — Both

FAILURES & SYSTEMS

- Climb, demo managed CLB CSTR; PROG page review — A
- Cruise, A/P and A/THR logic — A
- FMGC exercises: holding, DIR TO, FIX INFO, intercept — A

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTCC / VTBS
ALTERNATE	VTBD
FLIGHT NUMBER	THA2
COST INDEX	30
PLANNED CRUISE	FL320
FLAPS	1 or 2
ZFW (T)	48
ZFWCG (%MAC)	30.5
TAKE-OFF FUEL (T)	9

FIELD	BOTH TRAINEES
TOW (T)	57
TRIP FUEL (T)	2.7
TAKE-OFF CG (%MAC)	28.7
ROUTE	VTCCR36 PANTA2N PANTA Y7 BLAFF NORTA1C VTBSR19

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTCC
RUNWAY	18
WIND	170/4
VISIBILITY	10 km
CEILING	—
TEMP °C	30
QNH	1009
RWY COND.	Dry
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	180/10
VISIBILITY	10 km
CEILING	—
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Gate position	A only
2	Preflight procedures, transit stop, EXT PWR available	
3	Engine start	
4	Taxiing - before taxi procedures, after start flow patterns	
5	Taxiing - procedures	
6	Before take-off checks, flow patterns and procedures	
7	Take-off - instrument, packs OFF	A only
8	Follow SID, climb following SID altitude/FL constraints	
9	Climb, FL190	
10	Climb, demo managed CLB CSTR; PROG page review	A only
11	Cruise, A/P and A/THR logic	A only
12	FMGC exercises: holding, DIR TO, FIX INFO, intercept	A only
13	Descend for ILS approach	
14	Follow STAR	
15	Approach - ILS, with AP	Both
16	Landing - normal, autoland	Both
17	After landing procedures - shutdown	A only

TRAINEE B

#	TOPIC	A/B
1	Preflight procedures, transit stop, EXT PWR available	
2	Engine start	
3	Taxiing – before taxi procedures, after start flow patterns	
4	Taxiing – procedures	
5	Before take-off checks, flow patterns and procedures	
6	Take-off – instrument, packs ON	B only
7	Follow SID, climb following SID altitude/FL constraints	
8	Climb, FL190	
9	Cruise	B only

#	TOPIC	A/B
10	Descend, for an ILS approach	
11	Follow STAR	
12	Approach – ILS, with AP	**Both**
13	Landing – normal, autoland	**Both**

SESSION 2 OF 18

MFTD/FBS 3

A VTBS / VTBS | B VTBS / VTBS

Combined practice 13 items

TAKEOFF & DEPARTURE

Engine start with air start unit and electrical	A
Engine start, second engine using crossbleed	A
Take-off - instrument, packs OFF	Both

APPROACHES & GO-AROUNDS

Approach - 3D (RNP), with AP, using FINAL APP B specifies LNAV/VNAV minima.	Both
Go-around during instrument approach	Both
Approach - ILS, with AP	Both

LANDING

Landing - normal, autoland	Both
LOSS OF BRAKING, if applicable simulate during landing	B

FAILURES & SYSTEMS

Preflight procedures, transit; MEL APU inoperative	A
Climb, demo of mode reversions	A
Descend, FMGC exercises; demonstrate constraint symbols	Both
Demo of ground speed mini	B

FLIGHT HANDLING

Holding procedures	Both

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTBS
ALTERNATE	VTBD
FLIGHT NUMBER	THA3
COST INDEX	30
PLANNED CRUISE	—
FLAPS	1 or 2
ZFW (T)	52
ZFWCG (%MAC)	28.6
TAKE-OFF FUEL (T)	9
TOW (T)	61
TRIP FUEL (T)	—
TAKE-OFF CG (%MAC)	27.3
ROUTE	—

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	Calm
VISIBILITY	CAVOK
CEILING	—
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19

FIELD	BOTH TRAINEES
WIND	Calm
VISIBILITY	CAVOK
CEILING	—
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	Preflight procedures, transit; MEL APU inoperative	A only
2	Engine start with air start unit and electrical	A only
3	Engine start, second engine using crossbleed	A only
4	Taxiing, before take-off flow patterns and procedures	
5	Take-off - instrument, packs OFF	Both
6	Follow SID, climb following SID altitude/FL constraints	
7	Holding procedures	Both
8	Climb, demo of mode reversions	A only
9	Cruise	
10	Descend, FMGC exercises; demonstrate constraint symbols	Both
11	Follow ATC instructions; radar vectors for RNP approach	A only
12	Approach - 3D (RNP), with AP, using FINAL APP	Both
13	Go-around during instrument approach	Both
14	Radar vectoring for ILS approach	
15	Approach - ILS, with AP	Both
16	Landing - normal, autoland	Both
17	After landing procedures - shutdown	A only

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off position	B only

#	TOPIC	A/B
2	Taxiing, before take-off flow patterns and procedures	
3	Take-off – instrument, packs OFF	**Both**
4	Follow SID, climb following SID altitude/FL constraints	
5	Holding procedures	**Both**
6	Climb, FMGC exercises, demonstrate constraint symbols	**Both**
7	Cruise	
8	Descend	**B only**
9	Follow ATC instructions, for RNP approach	**B only**
10	Approach – 3D (RNP), with AP, to LNAV/VNAV minima, using FINAL APP	**Both**
11	Go-around – during instrument approach	**Both**
12	Radar vectoring, for ILS approach	
13	Demo of ground speed mini	**B only**
14	Approach – ILS, with AP	**Both**
15	LOSS OF BRAKING, if applicable simulate during landing	**B only**
16	Landing – normal, autoland	**Both**
17	After landing procedures	**B only**

SESSION 3 OF 18

MFTD/FBS 4

A VTBS / VTSP | B VTBS / VTSP

Combined practice 19 items

TAKEOFF & DEPARTURE

Take-off - instrument	Both
Engine start, manual engine start	B

APPROACHES & GO-AROUNDS

Approach - 2D VOR, down to MDH/A, using FINAL APP	A
Go-around during instrument approach	Both B ×2
Approach - 2D VOR, down to MDH/A, using NAV/FPA; freeze at minima and reposition	A
Approach - 3D ILS, with AP	Both
Approach – 2D (VOR), down to MDH/A, VOR	B
Approach – 2D, down to MDH/A, VOR, using TRK/FPA, freeze at minima and reposition	B

LANDING

Landing - normal, autoland	A

FAILURES & SYSTEMS

FUEL L(R) TK PUMP 1(2) LO PR	A
FUEL L(R) TK PUMP 1+2 LO PR	A
Gravity fuel feeding	A
ENG 1(2) FAIL, no damage	A
Engine relight in flight, successful relight	A
ELEC IDG 1(2) OIL LO PR/OVHT	A
FMGC exercises: position monitor, navaid, stored navaid, SEC F-PLN new DEST for immediate return or T/O ALT	B
PROG, F-PLN and PERF page review	B

FPV (bird) and FPD demonstration	B
CAB PR SYS 1(2) FAULT, ECAM introduction	B

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTSP
ALTERNATE	VTSS
FLIGHT NUMBER	THA4
COST INDEX	30
PLANNED CRUISE	FL340
FLAPS	1 or 2
ZFW (T)	54
ZFWCG (%MAC)	32.7
TAKE-OFF FUEL (T)	9
TOW (T)	63
TRIP FUEL (T)	3.2
TAKE-OFF CG (%MAC)	30.7
ROUTE	VTBSR19 VANKO1J VANKO Y8 SAVSA SAVSA1D VTSPR27

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	240/10
VISIBILITY	7 km
CEILING	—
TEMP °C	30
QNH	1007
RWY COND.	Dry
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTSP
RUNWAY	27
WIND	240/10
VISIBILITY	7 km
CEILING	—
TEMP °C	30
QNH	1007
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Take-off position	A only
2	Before take-off checks, flow patterns and procedures	
3	Take-off - instrument	Both
4	Follow SID, climb SID altitude/FL	
5	Radar vectoring for VOR/DME approach	
6	Approach - 2D VOR, down to MDH/A, using FINAL APP	A only
7	Go-around during instrument approach	Both
8	FUEL L(R) TK PUMP 1(2) LO PR	A only
9	FUEL L(R) TK PUMP 1+2 LO PR	A only
10	Gravity fuel feeding	A only
11	ENG 1(2) FAIL, no damage	A only
12	Engine relight in flight, successful relight	A only
13	Radar vectoring for VOR/DME approach	
14	ELEC IDG 1(2) OIL LO PR/OVHT	A only
15	Approach - 2D VOR, down to MDH/A, using NAV/FPA; freeze at minima and reposition	A only
16	Approach - 3D ILS, with AP	Both
17	Landing - normal, autoland	A only

#	TOPIC	A/B
18	After landing procedures - shutdown	

TRAINEE B

#	TOPIC	A/B
1	Preflight procedures, EXT PWR available	B only
2	FMGC exercises: position monitor, navaid, stored navaid, SEC F-PLN new DEST for immediate return or T/O ALT	B only
3	PROG, F-PLN and PERF page review	B only
4	Engine start, manual engine start	B only
5	Taxiing – procedures	B only
6	Before take-off checks, flow patterns and procedures	
7	Take-off – instrument	Both
8	Follow SID, climb SID altitude/FL	
9	Radar vectoring, demo of FPV (bird) and FPD	B only
10	Approach – ILS, with AP	Both
11	Go-around during instrument approach	Both
12	Radar vectoring, for VOR/DME approach	
13	Approach – 2D (VOR), down to MDH/A, VOR	B only
14	Go-around during instrument approach	Both
15	CAB PR SYS 1(2) FAULT, ECAM introduction	B only
16	Radar vectoring, for same VOR/DME as above	B only
17	Approach – 2D, down to MDH/A, VOR, using TRK/FPA, freeze at minima and reposition	B only
18	After landing procedures – shutdown	

SESSION 4 OF 18

MFTD/FBS 5

A VTBS / VTSP | B VTBS / VTSP

Combined practice 17 items

TAKEOFF & DEPARTURE

START FAULT, hot start	A
Take-off - instrument	Both

APPROACHES & GO-AROUNDS

Approach - 2D LOC, down to MDH/A, using LOC/FPA	Both A ×2 B ×2
Go-around during instrument approach	Both
Go-around during instrument approach at 500 ft AGL	Both
Approach - ILS, with AP	Both

LANDING

Landing - normal, autoland	Both
LOSS OF BRAKING, discuss	B

FAILURES & SYSTEMS

SLATS FAULT/LOCKED before extension, SLATS <1	Both
Emergency evacuation procedure review	A
AFT/BULK CRG SMOKE	A
Emergency evacuation	A
ENG 1(2) FAIL, no damage	B
Engine relight in flight, successful relight	B
HYD Y RSVR LO LVL	B

FLIGHT HANDLING

Holding procedures for LOC/DME approach	A

Holding procedures; restore malfunction after missed approach altitude — Both

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTSP
ALTERNATE	VTSG
FLIGHT NUMBER	THA5
COST INDEX	30
PLANNED CRUISE	FL340
FLAPS	1 or 2
ZFW (T)	52
ZFWCG (%MAC)	27.3
TAKE-OFF FUEL (T)	10
TOW (T)	62
TRIP FUEL (T)	3.2
TAKE-OFF CG (%MAC)	30
ROUTE	VTBSR19 VANKO1J VANKO Y8 SAVSA SAVSA1D VTSPR27

Flight conditions

DEPARTURE

FIELD	A	B
AIRPORT	VTBS	VTBS
RUNWAY	19	19
WIND	280/10	280/10
VISIBILITY DIFFERENT	5 km RA	5 km
CEILING	OVC005	OVC005
TEMP °C DIFFERENT	25	22
QNH	1003	1003
RWY COND.	Wet	Wet
ANTI-ICE	Off	Off

ARRIVAL

FIELD	A	B
AIRPORT	VTSP	VTSP
RUNWAY	27	27
WIND	300/10	300/10
VISIBILITY DIFFERENT	5 km RA	5 km
CEILING	OVC005	OVC005
TEMP °C	22	22
QNH DIFFERENT	1005	1003
RWY COND.	Wet	Wet
ANTI-ICE	Off	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	Preflight procedures, transit stop, EXT PWR available	A only
2	START FAULT, hot start	A only
3	Taxiing - procedures	A only
4	Before take-off checks, flow patterns and procedures	
5	Take-off - instrument	Both
6	Follow SID, climb SID altitude/FL	
7	Radar vectoring for LOC/DME approach	
8	Approach - 2D LOC, down to MDH/A, using LOC/FPA	Both
9	Go-around during instrument approach	Both
10	Holding procedures for LOC/DME approach	A only
11	SLATS FAULT/LOCKED before extension, SLATS <1	Both
12	Approach - 2D LOC, down to MDH/A, using LOC/FPA	Both
13	Go-around during instrument approach at 500 ft AGL	Both
14	Holding procedures; restore malfunction after missed approach altitude	Both
15	Radar vectoring for ILS; review emergency evacuation procedure	A only
16	AFT/BULK CRG SMOKE	A only
17	Approach - ILS, with AP	Both

#	TOPIC	A/B
18	Landing - normal, autoland	Both
19	Emergency evacuation	A only

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off position	B only
2	Before take-off checks, flow patterns and procedures	
3	Take-off – instrument	Both
4	Follow SID, climb SID altitude/FL	
5	Radar vectoring, for LOC/DME approach	
6	Approach – 2D, down to MDH/A, LOC, using LOC/FPA	Both
7	Go-around – during instrument approach	Both
8	SLATS FAULT/LOCKED, select fault before extension, SLATS < 1	Both
9	Radar vectoring, for LOC/DME approach	
10	Approach – 2D, down to MDH/A, LOC, using LOC/FPA	Both
11	Go-around – during instrument approach, at 500 ft AGL	Both
12	Holding procedures, restore malfunction after reaching missed approach altitude	Both
13	ENG 1(2) FAIL, no damage	B only
14	Engine relight in flight, successful relight	B only
15	Radar vectoring, for ILS approach	B only
16	HYD Y RSVR LO LVL	B only
17	Approach – 3D ILS, with AP	Both
18	Landing – normal, autoland	Both
19	LOSS OF BRAKING, discuss	B only
20	After landing procedures – shutdown	B only

SESSION 5 OF 18

MFTD/FBS 6

A VTSP / VTBS | B VTSP / VTBS

Combined practice 10 items

TAKEOFF & DEPARTURE

Take-off - instrument	Both A ×2
START FAULT, start valve engine 2	B

APPROACHES & GO-AROUNDS

Approach - ILS, with AP	Both A ×2

LANDING

Landing - normal, autoland	Both A ×2

FAILURES & SYSTEMS

UNRELIABLE SPEED INDICATION and turbulent air demo	A
Cabin pressure failure / emergency descent, no damage; restore below FL100 or MSA	Both B ×2
ADR 1+2 (1+3) (2+3) FAULT demo; restore systems	A
Fuel leak	A
Air conditioning smoke	B
Removal of smoke/fumes	B

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTSP / VTBS
ALTERNATE	VTBD
FLIGHT NUMBER	THA6

FIELD	BOTH TRAINEES
COST INDEX	30
PLANNED CRUISE	FL330
FLAPS	1 or 2
ZFW (T)	48
ZFWCG (%MAC)	30.5
TAKE-OFF FUEL (T)	10
TOW (T)	58
TRIP FUEL (T)	3.1
TAKE-OFF CG (%MAC)	28.7
ROUTE	VTSPR27 EMRIT1B EMRIT Y99 NONEL Y98 LEBIM LEBIM1C VTBSR19

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTSP
RUNWAY	27
WIND	260/15
VISIBILITY	CAVOK
CEILING	—
TEMP °C	30
QNH	1008
RWY COND.	Dry
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	250/10
VISIBILITY	CAVOK
CEILING	—
TEMP °C	30

FIELD	BOTH TRAINEES
QNH	1006
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Take-off position	A only
2	Before take-off checks, flow patterns and procedures	
3	Take-off - instrument	Both
4	Follow SID, climb SID altitude/FL	
5	UNRELIABLE SPEED INDICATION and turbulent air demo	A only
6	INIT: FL350	A only
7	Cabin pressure failure / emergency descent, no damage; restore below FL100 or MSA	Both
8	ADR 1+2 (1+3) (2+3) FAULT demo; restore systems	A only
9	Radar vectoring for ILS approach	
10	Approach - ILS, with AP	Both
11	Landing - normal, autoland	Both
12	Take-off - instrument	Both
13	Follow SID, climb SID altitude/FL	
14	Fuel leak	A only
15	Radar vectoring for ILS approach	
16	Approach - ILS, with AP	Both
17	Landing - normal, autoland	Both
18	After landing procedures - shutdown	A only

TRAINEE B

#	TOPIC	A/B
1	Preflight procedures, transit stop, EXT PWR available	B only
2	START FAULT, start valve engine 2	B only
3	Taxiing – procedures	B only
4	Before take-off checks, flow patterns and procedures	

#	TOPIC	A/B
5	Take-off – instrument	**Both**
6	Follow SID, climb SID altitude/FL	
7	INIT: FL290	**B only**
8	Air conditioning smoke	**B only**
9	Removal of smoke/fumes; when completed, reposition FL350	**B only**
10	Cabin pressure failure / emergency descent, no damage	**Both**
11	Restore all systems after level-off below FL100 or MSA	**Both**
12	Radar vectoring, for ILS approach	
13	Approach – ILS, with AP	**Both**
14	Landing – normal, autoland	**Both**

SESSION 6 OF 18

MFTD/FBS 7

A VTBS / VTSP | B VTBS / VTSP

Combined practice 13 items

TAKEOFF & DEPARTURE

Take-off - instrument	A
Take-off - engine failure after reaching V2, repeat to proficiency	Both

APPROACHES & GO-AROUNDS

Approach - 3D (RNP), with AP, to LNAV/VNAV minima, using FINAL APP	Both
Go-around at minima, one engine inoperative	Both
Approach - ILS, with AP, one engine inoperative	Both A ×2
Go-around – at minima, engine failure at go-around	B
Approach – 2D, down to MDH/A, LOC, one engine inoperative	B

LANDING

Landing - one engine inoperative, autoland	Both A ×2

FAILURES & SYSTEMS

ENG 1(2) FAIL, no damage; drift down using standard strategy	Both B ×2
Engine relight in flight; restore all systems	Both
ENG 1(2) FIRE, extinguishable after 1st bottle	B
APU FIRE	B
Emergency evacuation	B

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTSP
ALTERNATE	—
FLIGHT NUMBER	THA7
COST INDEX	30
PLANNED CRUISE	FL320
FLAPS	1 or 2
ZFW (T)	46
ZFWCG (%MAC)	35
TAKE-OFF FUEL (T)	10
TOW (T)	56
TRIP FUEL (T)	3.2
TAKE-OFF CG (%MAC)	32.4
ROUTE	VTBSR19 VANKO1J VANKO Y8 SAVSA SAVSA1D VTSPR27

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	060/6
VISIBILITY	CAVOK
CEILING	—
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTSP
RUNWAY	27

FIELD	BOTH TRAINEES
WIND	150/5
VISIBILITY	CAVOK
CEILING	—
TEMP °C	27
QNH	1007
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	Preflight procedures, transit stop, EXT PWR available	A only
2	Engine start	A only
3	Taxiing - procedures	A only
4	Before take-off checks, flow patterns and procedures	
5	Take-off - instrument	A only
6	Follow SID, climb SID altitude/FL	A only
7	INIT: FL350	
8	ENG 1(2) FAIL, no damage; drift down using standard strategy	Both
9	Engine relight in flight; restore all systems	Both
10	Reposition for RNP approach	
11	Approach - 3D (RNP), with AP, to LNAV/VNAV minima, using FINAL APP	Both
12	Go-around at minima, one engine inoperative	Both
13	Radar vectoring for ILS approach	
14	Approach - ILS, with AP, one engine inoperative	Both
15	Landing - one engine inoperative, autoland	Both
16	Take-off - engine failure after reaching V2, repeat to proficiency	Both
17	Approach - 3D ILS, with AP, one engine inoperative	Both
18	Landing - one engine inoperative, autoland	Both

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off position	**B only**
2	Before take-off checks, flow patterns and procedures	
3	Take-off – engine failure after reaching V2, repeat to proficiency	**Both**
4	INIT: FL350	
5	ENG 1(2) FAIL, no damage	**Both**
6	Drift down using standard strategy	**Both**
7	Engine relight in flight, restore all systems	**Both**
8	Reposition for RNP approach	
9	Approach – 3D (RNP), with AP, to LNAV/VNAV minima, using FINAL APP	**Both**
10	Go-around – at minima, engine failure at go-around	**B only**
11	ENG 1(2) FIRE, extinguishable after 1st bottle	**B only**
12	Radar vectoring, for LOC/DME approach	**B only**
13	Approach – 2D, down to MDH/A, LOC, one engine inoperative	**B only**
14	Go-around – at minima, one engine inoperative	**Both**
15	Radar vectoring, for ILS approach	
16	Approach – 3D ILS, with AP, one engine inoperative	**Both**
17	Landing – one engine inoperative, autoland	**Both**
18	APU FIRE	**B only**
19	Emergency evacuation	**B only**

SESSION 7 OF 18

MFTD/FBS 8

A VTSG / VTBS | B VTSG / VTBS

Combined practice 19 items

TAKEOFF & DEPARTURE

Take-off - engine failure after reaching V2	A
Rejected take-off - engine fire below V1, not extinguishable	A
Take-off – instrument	B

APPROACHES & GO-AROUNDS

2D LOC approach to MDH/A A specifies LOC/FPA; B does not specify a guidance mode.	Both
Go-around at minima, one engine inoperative	Both A ×2
Approach - ILS, with AP, one engine inoperative	Both A ×2 B ×2
Go-around – at minima, engine failure at go-around; one engine inoperative	B

LANDING

Landing - one engine inoperative, autoland	Both

FAILURES & SYSTEMS

Engine relight in flight	A
ANTI ICE STBY PITOT or L(R) STAT or AOA demo	A
Emergency evacuation	A
FMGC 1(2) fault – demo FMGC 1 fail, demo FMGC 2 fail	B
FMGC 1+2 fault, reset using QRH System Reset table	B
Demo SDAC 1+2 fault and reset	B
Demo FWC 1+2 fault and reset	B
Compare QRH Unreliable Speed with QRH Severe Turbulence table	B

ELEC EMER CONFIG, demo B

ENG 1(2) FIRE, extinguishable after 2nd bottle B

FLIGHT HANDLING

Holding procedures, FMGC review B

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTSG / VTBS
ALTERNATE	VTBD
FLIGHT NUMBER	THA8
COST INDEX	30
PLANNED CRUISE	FL330
FLAPS	1 or 2
ZFW (T)	55
ZFWCG (%MAC)	28.6
TAKE-OFF FUEL (T)	10
TOW (T)	65
TRIP FUEL (T)	2.7
TAKE-OFF CG (%MAC)	27.3
ROUTE	VTSGR32 OSPEX1F OSPEX Y99 NONEL Y98 LEBIM LEBIM1C VTBSR19

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTSG
RUNWAY	32
WIND	270/10
VISIBILITY	7 km
CEILING	2000
TEMP °C	30
QNH	1008

FIELD	BOTH TRAINEES
RWY COND.	Dry
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	170/10
VISIBILITY	7 km
CEILING	2000
TEMP °C	30
QNH	1008
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Take-off position, quick set-up	
2	Take-off - engine failure after reaching V2	A only
3	Engine relight in flight	A only
4	Radar vectoring for LOC/DME approach	A only
5	ANTI ICE STBY PITOT or L(R) STAT or AOA demo	A only
6	Approach - 2D LOC, down to MDH/A, using LOC/FPA	Both
7	Go-around at minima, one engine inoperative	Both
8	Radar vectoring for ILS approach	
9	Approach - ILS, with AP, one engine inoperative	Both
10	Go-around at minima, one engine inoperative; follow missed approach procedure	Both
11	Radar vectoring for ILS approach	
12	Approach - 3D ILS, with AP, one engine inoperative	Both
13	Landing - one engine inoperative, autoland	Both
14	Rejected take-off - engine fire below V1, not extinguishable	A only

#	TOPIC	A/B
15	Emergency evacuation	A only

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off position, quick set-up	
2	Take-off – instrument	B only
3	Follow SID, climb SID altitude/FL	B only
4	Holding procedures, FMGC review	B only
5	FMGC 1(2) fault – demo FMGC 1 fail, demo FMGC 2 fail	B only
6	FMGC 1+2 fault, reset using QRH System Reset table	B only
7	Demo SDAC 1+2 fault and reset	B only
8	Demo FWC 1+2 fault and reset	B only
9	INIT: FL310	B only
10	Compare QRH Unreliable Speed with QRH Severe Turbulence table	B only
11	ELEC EMER CONFIG, demo	B only
12	INIT: FL250	B only
13	Approach – 2D, down to MDH/A, LOC	Both
14	Go-around – at minima, engine failure at go-around; one engine inoperative	B only
15	Radar vectoring, for ILS approach	
16	ENG 1(2) FIRE, extinguishable after 2nd bottle	B only
17	Approach – ILS, with AP, one engine inoperative	Both
18	Go-around – at minima, one engine inoperative	Both
19	Radar vectoring, for ILS approach	
20	Approach – ILS, with AP, one engine inoperative	Both
21	Landing – one engine inoperative, autoland	Both

SESSION 8 OF 18

MFTD/FBS 9

A VTBS / VHHH | B VHHH / VTBS

Combined practice 19 items

TAKEOFF & DEPARTURE

START FAULT, hot start	A
Engine start; manual engine start	A
Instrument takeoff A specifies packs OFF; B does not specify a packs setting.	Both
Rejected take-off - engine fire below V1, not extinguishable	A

APPROACHES & GO-AROUNDS

Approach - 2D (LOC/DME), down to MDH/A	A
Go-around at minima, one engine inoperative	A
Approach - ILS, with AP, one engine inoperative	Both
Approach – 2D, down to MDH/A, VOR, using TRK/FPA	B
Discontinued approach, due to system malfunctions; remove landing gear malfunction after go-around	B

LANDING

Overweight landing procedures with LOC/DME vectors	A
Landing - one engine inoperative, autoland	A
Landing – one engine inoperative, contaminated RWY, autoland	B

FAILURES & SYSTEMS

ENG 1(2) FIRE; drift down using standard strategy	A
Emergency evacuation	A
Performance calculation, contaminated runway	B
Cabin pressure failure / emergency descent, no damage	B
GEAR NOT DOWNLOCKED	B

ENG 1(2) FAIL, with damage B

FLIGHT HANDLING

Holding procedures A

Flight setup

FIELD	TRAINEE A	TRAINEE B
FROM / TO DIFFERENT	VTBS / VHHH	VHHH / VTBS
ALTERNATE DIFFERENT	VMMC	VTBD
FLIGHT NUMBER	THA9	THA9
COST INDEX	30	30
PLANNED CRUISE DIFFERENT	FL330	FL340
FLAPS	1 or 2	1 or 2
ZFW (T)	56	56
ZFWCG (%MAC)	25.8	25.8
TAKE-OFF FUEL (T)	13	13
TOW (T)	69	69
TRIP FUEL (T) DIFFERENT	5.9	7.5
TAKE-OFF CG (%MAC)	25	25
ROUTE DIFFERENT	VTBSR19 SELKA1J SELKA A1 TOPER W42 PAKRI DCT GRASO Y15 GUROK A202 SIKOU V571 CANTO CANTO3A VHHHR07C	VHHHR25L PECAN2B PECAN V10 SIKOU A202 SAV B460 ROT R345 GRASO Y20 KRT DCT RUKSA EASTE1C VTBSR19

Flight conditions

DEPARTURE

FIELD	A	B
AIRPORT DIFFERENT	VTBS	VHHH
RUNWAY DIFFERENT	19	25L
WIND DIFFERENT	180/10	330/10
VISIBILITY DIFFERENT	5 km	3 km
CEILING	2000	2000
TEMP °C DIFFERENT	30	25

FIELD	A	B
QNH DIFFERENT	1008	1012
RWY COND.	Dry	Dry
ANTI-ICE	Off	Off

ARRIVAL

FIELD	A	B
AIRPORT DIFFERENT	VHHH	VTBS
RUNWAY DIFFERENT	07C	19
WIND DIFFERENT	360/10	270/10
VISIBILITY	5 km	5 km
CEILING DIFFERENT	1000	5000
TEMP °C DIFFERENT	24	30
QNH DIFFERENT	1010	1007
RWY COND.	Dry	Dry
ANTI-ICE	Off	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	Procedures, no EXT PWR available	A only
2	Performance calculation	A only
3	START FAULT, hot start	A only
4	Engine start; manual engine start	A only
5	Taxiing - procedures	
6	Before take-off checks, flow patterns and procedures	
7	Take-off - instrument, packs OFF	Both
8	Follow SID, climb SID altitude/FL	
9	Holding procedures	A only
10	INIT: FL290	A only
11	ENG 1(2) FIRE; drift down using standard strategy	A only
12	Radar vectoring for LOC/DME; overweight landing procedures	A only
13	Approach - 2D (LOC/DME), down to MDH/A	A only

#	TOPIC	A/B
14	Go-around at minima, one engine inoperative	A only
15	Radar vectoring for ILS approach	
16	Approach - ILS, with AP, one engine inoperative	Both
17	Landing - one engine inoperative, autoland	A only
18	Rejected take-off - engine fire below V1, not extinguishable	A only
19	Emergency evacuation	A only

TRAINEE B

#	TOPIC	A/B
1	Preflight procedures, EXT PWR available	B only
2	Performance calculation, contaminated runway	B only
3	Engine start	B only
4	Taxiing – procedures	
5	Before take-off checks, flow patterns and procedures	
6	Take-off – instrument	Both
7	Follow SID, climb SID altitude/FL	
8	INIT: FL310	B only
9	Cabin pressure failure / emergency descent, no damage	B only
10	Radar vectoring, for VOR/DME approach	B only
11	Approach – 2D, down to MDH/A, VOR, using TRK/FPA	B only
12	GEAR NOT DOWNLOCKED	B only
13	Discontinued approach, due to system malfunctions; remove landing gear malfunction after go-around	B only
14	ENG 1(2) FAIL, with damage	B only
15	Radar vectoring, for ILS approach	
16	Approach – ILS, with AP, one engine inoperative	Both
17	Landing – one engine inoperative, contaminated RWY, autoland	B only
18	After landing procedures – shutdown	B only

SESSION 9 OF 18

FFS 1

A VTBS / VTBS | B VTBS / VTBS

Combined practice 17 items

TAKEOFF & DEPARTURE

Take-off - normal, CONF 2	Both

APPROACHES & GO-AROUNDS

Approach - ILS, with AP	Both

LANDING

Landing - normal, autoland	Both
Manual landing practice ×2	Both

FAILURES & SYSTEMS

Preflight procedures; introduction of RMP/ACP	A
Demonstrate AP and FD connect/disconnect logic	Both
Acceleration / deceleration, trim changes, auto-trim demo	Both
Demonstrate sidestick priority logic and disconnect	B

FLIGHT HANDLING

Manual flight - with FD	Both
Manual flight - without FD; introduce FPV (bird)	Both
Manual flight - slow speed, low altitude	Both
Turns - with spoilers	Both
Turns - without spoilers	Both
Steep turns, 45° bank, left and right	Both
UPRT recovery from nose-high, Normal law and protections	Both
UPRT recovery from nose-low, Normal law and protections	Both

Manual flight - normal speed, high altitude **Both**

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTBS
ALTERNATE	VTBD
FLIGHT NUMBER	THA01
COST INDEX	30
PLANNED CRUISE	FL320
FLAPS	1 or 2
ZFW (T)	49.5
ZFWCG (%MAC)	32
TAKE-OFF FUEL (T)	8
TOW (T)	57.5
TRIP FUEL (T)	—
TAKE-OFF CG (%MAC)	30
ROUTE	—

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	280/10
VISIBILITY	5 km
CEILING	2000
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	290/15
VISIBILITY	3 km
CEILING	1000
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	Preflight procedures; introduction of RMP/ACP	A only
2	Performance calculation	A only
3	Use of checklist before starting engine	A only
4	Taxiing	A only
5	Before take-off checks	
6	Take-off - normal, CONF 2	Both
7	Follow ATC instructions, climb 12,000 ft/FL120; perform FL100 checks	
8	Demonstrate AP and FD connect/disconnect logic	Both
9	Acceleration / deceleration, trim changes, auto-trim demo	Both
10	Manual flight - with FD	Both
11	Manual flight - without FD; introduce FPV (bird)	Both
12	Manual flight - slow speed, low altitude	Both
13	Turns - with spoilers	Both
14	Turns - without spoilers	Both
15	Steep turns, 45° bank, left and right	Both
16	UPRT recovery from nose-high, Normal law and protections	Both
17	UPRT recovery from nose-low, Normal law and protections	Both

#	TOPIC	A/B
18	Manual flight - normal speed, high altitude	Both
19	Reposition for ILS approach	A only
20	Approach - ILS, with AP	Both
21	Landing - normal, autoland	Both
22	Reposition to final for manual landing practice x2	Both

TRAINEE B

#	TOPIC	A/B
1	Before take-off checks	
2	Take-off – normal, CONF 2	Both
3	Follow ATC instructions, climb 12,000 ft/FL120; perform 10,000 ft/FL100 checks	
4	Demonstrate AP and FD connect/disconnect logic	Both
5	Demonstrate sidestick priority logic and disconnect	B only
6	Acceleration / deceleration, trim changes, auto-trim demo	Both
7	Manual flight – with FD	Both
8	Manual flight – without FD, introduce the FPV (bird)	Both
9	Manual flight – slow speed, low altitude	Both
10	Turns – with spoilers	Both
11	Turns – without spoilers	Both
12	Steep turns with 45° bank, 180° to 360°, left and right	Both
13	UPRT recovery from nose-high at various bank angles, Normal law exercise and protections demo	Both
14	UPRT recovery from nose-low at various bank angles, Normal law exercise and protection demo	Both
15	Manual flight – normal speed, high altitude	Both
16	Follow ATC instructions, reposition for the ILS approach	B only
17	Approach – 3D (ILS), with AP	Both
18	Landing – normal, autoland; manual landing practice ×2	Both
19	Reposition to final for manual landing procedures	B only
20	Taxiing in, after landing procedures	B only
21	ENG shutdown, parking procedures	B only

SESSION 10 OF 18

FFS 2

A VTBS / VTCC | B VTBS / VTCC

Combined practice 18 items

TAKEOFF & DEPARTURE

Take-off - normal	Both
Take-off - crosswind, CONF 1+F	Both

APPROACHES & GO-AROUNDS

Approach - visual	Both
Approach - 3D (RNP), with AP, LNAV/VNAV minima, FINAL APP	A
Go-around - rejected landing below DH/MDH	Both
Approach - 2D (RNP), LNAV minima, NAV/FPA	Both
Go-around - rejected landing after touchdown	Both

LANDING

Landing - normal	Both
Landing - crosswind, CONF 3	Both

FAILURES & SYSTEMS

TCAS RA event	Both
Procedural instrument flying and manoeuvring A combines this with RNAV STAR or radar vectors; B lists that setup separately.	Both
180° turn on runway	Both

FLIGHT HANDLING

Approach-to-stall exercises B also covers stall warning or stall avoidance devices.	Both
Recovery from stall - clean configuration, low altitude	Both
Recovery from stall - clean configuration, near maximum operating altitude	Both

Visual circuits	Both A ×2 B ×2
Recovery from stall - landing configuration	Both
Recovery from stall - take-off configuration	Both

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTCC
ALTERNATE	VTCT
FLIGHT NUMBER	THA02
COST INDEX	30
PLANNED CRUISE	FL330
FLAPS	1 or 2
ZFW (T)	49.5
ZFWCG (%MAC)	32
TAKE-OFF FUEL (T)	8
TOW (T)	57.5
TRIP FUEL (T)	2.8
TAKE-OFF CG (%MAC)	30
ROUTE	VTBSR19 OLVUK1J OLVUK Y26 MARNI MARNI2A VTCCR36

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	VRB/05
VISIBILITY	4 km LT RA
CEILING	1000
TEMP °C	25
QNH	1003
RWY COND.	Wet

FIELD	BOTH TRAINEES
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTCC
RUNWAY	36
WIND	300/10
VISIBILITY	5 km RA
CEILING	1000
TEMP °C	25
QNH	1004
RWY COND.	Wet
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	Take-off - normal	Both
2	Follow SID, RNAV	
3	TCAS RA event	Both
4	Approach-to-stall exercises	Both
5	Recovery from stall - clean configuration, low altitude	Both
6	Recovery from stall - clean configuration, near maximum operating altitude	Both
7	Reposition for visual circuit	A only
8	Visual circuits	Both
9	Recovery from stall - landing configuration	Both
10	Approach - visual	Both
11	Landing - normal	Both
12	Take-off - crosswind, CONF 1+F	Both
13	Recovery from stall - take-off configuration	Both
14	Procedural instrument flying; RNAV STAR or radar vectors for RNP	Both
15	Approach - 3D (RNP), with AP, LNAV/VNAV minima, FINAL APP	A only

#	TOPIC	A/B
16	Go-around - rejected landing below DH/MDH	Both
17	Radar vectors for same 2D RNP, LNAV minima	A only
18	Approach - 2D (RNP), LNAV minima, NAV/FPA	Both
19	Go-around - rejected landing after touchdown	Both
20	Visual circuits	Both
21	Landing - crosswind, CONF 3	Both
22	180° turn on runway	Both
23	After landing procedures - shutdown	A only

TRAINEE B

#	TOPIC	A/B
1	Preflight procedures	B only
2	Performance calculation	B only
3	Use of checklist before starting engines	B only
4	Before take-off checks, during taxiing	B only
5	Take-off – normal	Both
6	Follow SID, RNAV	
7	TCAS RA event	Both
8	Stall warning devices or stall avoidance devices, approach-to-stall exercises	Both
9	Recovery from stall – clean configuration, low altitude	Both
10	Recovery from stall – clean configuration, near maximum operating altitude	Both
11	Follow ATC instructions, reposition for visual circuit	B only
12	Visual circuits	Both
13	Recovery from stall – landing configuration	Both
14	Approach – visual	Both
15	Landing – normal	Both
16	Take-off – crosswind, CONF 1+F	Both
17	Recovery from stall – take-off configuration	Both
18	Procedural instrument flying and manoeuvring	Both
19	RNAV STAR or radar vectoring for RNP approach using FINAL APP	B only
20	Go-around – rejected landing below DA/MDA	Both

#	TOPIC	A/B
21	Radar vectoring, for the same RNP approach as above, now as 2D (RNP), using LNAV minima	**B only**
22	Approach – 2D (RNP), down to LNAV minima, using NAV/FPA	**Both**
23	Go-around – rejected landing after touchdown	**Both**
24	Visual circuits	**Both**
25	Landing – crosswind, CONF 3	**Both**
26	Perform a 180° turn on RWY	**Both**

SESSION 11 OF 18

FFS 3

A VHHH / VTBS | B VHHH / VTBS

Combined practice 21 items

TAKEOFF & DEPARTURE

Take-off - instrument, CONF 3, contaminated runway	Both
Take-off - crosswind, night	Both
Windshear - take-off	Both

APPROACHES & GO-AROUNDS

Approach - 2D LOC, MDH/A, using LOC/FPA	Both
Go-around during instrument approach	A
Approach - circling following LOC/DME	Both
ILS approach with glide-slope interception from above B specifies a 200 ft DH/A.	Both
Go-around; other missed approach procedures	A
Discontinued approach	B
Go-around – at DH/A	B

LANDING

Landing - normal	Both
Windshear - landing B places landing windshear close to minima.	Both
Landing - crosswind 15-25 kt, braking action medium B adds more crosswind landing practice if time permits.	Both

FAILURES & SYSTEMS

ANTI ICE L(R) WINDOW; probe/window heat ON for cold weather	A
Use of engine anti-ice A specifies use during engine start; B lists use of engine anti-ice.	Both
Taxiing - slippery/contaminated taxiway	A

TCAS RA event	Both
EGPWS alerts	Both

FLIGHT HANDLING

UPRT recovery from nose-high at various bank angles	Both
UPRT recovery from nose-low at various bank angles	Both
Visual circuits, manual thrust, crosswind	Both

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VHHH / VTBS
ALTERNATE	VTBD
FLIGHT NUMBER	THA3
COST INDEX	30
PLANNED CRUISE	FL340
FLAPS	1 or 2
ZFW (T)	52
ZFWCG (%MAC)	28.6
TAKE-OFF FUEL (T)	13
TOW (T)	65
TRIP FUEL (T)	7.5
TAKE-OFF CG (%MAC)	27.3
ROUTE	VHHHR25L PECAN2B PECAN V10 SIKOU A202 SAV B460 ROT R345 GRASO Y20 KRT DCT RUKSA EASTE1C VTBSR19

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VHHH
RUNWAY	25L
WIND	VRB/05
VISIBILITY	3 km LT SN

FIELD	BOTH TRAINEES
CEILING	2000
TEMP °C	3
QNH	1025
RWY COND.	Contaminated
ANTI-ICE	On

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	180/10
VISIBILITY	5 km LT RA
CEILING	1000
TEMP °C	25
QNH	1010
RWY COND.	Wet
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	Preflight procedures, EXT PWR available	A only
2	ANTI ICE L(R) WINDOW; probe/window heat ON for cold weather	A only
3	Engine start, engine anti-ice	Both
4	Taxiing - slippery/contaminated taxiway	A only
5	Take-off - instrument, CONF 3, contaminated runway	Both
6	Follow SID, climb SID altitude/FL	A only
7	TCAS RA event	Both
8	UPRT recovery from nose-high at various bank angles	Both
9	UPRT recovery from nose-low at various bank angles	Both
10	Follow ATC instructions for LOC/DME approach	
11	Approach - 2D LOC, MDH/A, using LOC/FPA	Both

#	TOPIC	A/B
12	Go-around during instrument approach	A only
13	Radar vectoring for circling approach	
14	Approach - circling following LOC/DME	Both
15	Landing - normal	Both
16	Take-off - crosswind, night	Both
17	Windshear - take-off	Both
18	Set up for EGPWS exercise	A only
19	EGPWS alerts	Both
20	Approach - ILS, with AP, G/S interception from above	Both
21	Windshear - landing	Both
22	Go-around; other missed approach procedures	A only
23	Radar vectoring for visual circuit	
24	Visual circuits, manual thrust, crosswind	Both
25	Landing - crosswind 15-25 kt, braking action medium	Both

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off position	B only
2	Take-off – instrument, CONF 3, contaminated RWY	Both
3	Use of engine anti-ice	Both
4	TCAS RA event	Both
5	UPRT recovery from nose-high at various bank angles	Both
6	UPRT recovery from nose-low at various bank angles	Both
7	Follow ATC instructions, for LOC/DME approach	
8	Approach – 2D, down to MDH/A, LOC, using LOC/FPA	Both
9	Discontinued approach	B only
10	Radar vectoring, for circling approach	
11	Approach – circling, following a LOC/DME approach	Both
12	Landing – normal	Both
13	Take-off – crosswind, during night conditions	Both
14	Windshear – take-off	Both

#	TOPIC	A/B
15	Follow ATC instructions, set-up for EGPWS exercise	**B only**
16	EGPWS alerts	**Both**
17	ILS approach – to DH/A of 200 ft, G/S interception from above	**Both**
18	Windshear – landing, close to minima	**Both**
19	Go-around – at DH/A	**B only**
20	Radar vectoring, for visual circuit	
21	Visual circuits, using manual thrust, crosswind	**Both**
22	Landing – crosswind 15–25 kt, braking action medium; reposition to final for crosswind landing practice as time permits	**Both**
23	Taxiing – procedures, after landing	**B only**
24	After landing procedures – shutdown	**B only**

SESSION 12 OF 18

FFS 4

A VTBS / VTBS | B VTBS / VTBS

Combined practice 19 items

TAKEOFF & DEPARTURE

Take-off - max crosswind	Both
Windshear - take-off	Both

APPROACHES & GO-AROUNDS

Approach - ILS, with AP, using LOC/FPA	Both A ×2 B ×2
Go-around during instrument approach	Both

LANDING

Windshear - landing	Both
Landing - crosswind, in DIR LAW	Both

FAILURES & SYSTEMS

IR 1(2)(3) FAULT on PM side, no ECAM	A
IR 1+2 (1+3) (2+3) FAULT, no ECAM	A
F/CTL ALTN LAW and DIR LAW handling	Both
Mechanical backup demo	Both
Cabin pressure failure / emergency descent; structural damage suspected	A
RA 1+2 FAULT	Both
AFT/BULK CRG SMOKE	Both
Removal of smoke/fumes	Both
Emergency evacuation due to cargo fire	A
ADR 1(2)(3) FAULT	B
ADR 1+2 (1+3) (2+3) FAULT	B

Cabin pressure failure / emergency descent, no structural damage suspected B

FLIGHT HANDLING

Tuck under and Mach buffet in Normal and Alternate law Both

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTBS
ALTERNATE	VTBD
FLIGHT NUMBER	THA04
COST INDEX	30
PLANNED CRUISE	9,000 ft
FLAPS	1 or 2
ZFW (T)	54
ZFWCG (%MAC)	32.7
TAKE-OFF FUEL (T)	9
TOW (T)	63
TRIP FUEL (T)	—
TAKE-OFF CG (%MAC)	30.7
ROUTE	—

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	120/15
VISIBILITY	5 km
CEILING	1000
TEMP °C	28
QNH	1005
RWY COND.	Dry

FIELD	BOTH TRAINEES
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	110/20
VISIBILITY	3 km
CEILING	1000
TEMP °C	28
QNH	1005
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Take-off position	
2	Take-off - max crosswind	Both
3	Windshear - take-off	Both
4	Climb, 10,000 ft/FL100	
5	IR 1(2)(3) FAULT on PM side, no ECAM	A only
6	IR 1+2 (1+3) (2+3) FAULT, no ECAM	A only
7	F/CTL ALTN LAW and DIR LAW handling	Both
8	Mechanical backup demo	Both
9	Tuck under and Mach buffet, high-altitude handling	Both
10	Cabin pressure failure / emergency descent; structural damage suspected	A only
11	Radar vectoring for ILS approach	
12	RA 1+2 FAULT	Both
13	Approach - ILS, with AP, using LOC/FPA	Both
14	Windshear - landing	Both
15	Go-around during instrument approach	Both

#	TOPIC	A/B
16	AFT/BULK CRG SMOKE	Both
17	Removal of smoke/fumes	Both
18	Approach - ILS, with AP, using LOC/FPA	Both
19	Landing - crosswind, in DIR LAW	Both
20	Emergency evacuation due to cargo fire	A only

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off position	
2	Take-off – max crosswind	Both
3	Windshear – take-off	Both
4	Climb, 10,000 ft/FL100	
5	ADR 1(2)(3) FAULT	B only
6	ADR 1+2 (1+3) (2+3) FAULT	B only
7	F/CTL ALTN LAW and DIR LAW handling	Both
8	Mechanical backup, demo	Both
9	Tuck under and Mach buffets, aircraft handling at high altitude in Normal and Alternate law	Both
10	Cabin pressure failure / emergency descent, no structural damage suspected	B only
11	Radar vectoring, for ILS approach	
12	RA 1+2 FAULT	Both
13	Approach – ILS, with AP, using LOC/FPA	Both
14	Windshear – landing	Both
15	Go-around – during instrument approach	Both
16	AFT/BULK CRG SMOKE	Both
17	Removal of smoke/fumes	Both
18	Approach – ILS, with AP, using LOC/FPA	Both
19	Landing – crosswind, in DIR LAW	Both

SESSION 13 OF 18

FFS 5

A VTSG / VTSG | B VTSG / VTSG

Combined practice 19 items

TAKEOFF & DEPARTURE

Take-off - engine failure after reaching V2, familiarization	A
Take-off - engine failure between V1 and V2, repeat to proficiency	Both
Take-off - engine failure after reaching V2	Both
Take-off - crosswind	Both

APPROACHES & GO-AROUNDS

Approach - 3D RNP manually with FD, degradation or system failure	A
Go-around at minima, engine failure at go-around	A
Approach - ILS manually with FD, one engine inoperative	Both
Approach - ILS raw data, manually without FD	Both
Approach – ILS, manually, with FD	B
Go-around – during instrument approach	B
ENG 1(2) FAIL, during go-around	B

LANDING

Landing - one engine inoperative; repeat single-engine landing	Both
Landing - crosswind	Both

FAILURES & SYSTEMS

Engine relight in flight A repositions to FL350 after restart.	Both
ENG 1(2) FAIL, drift down procedure	A
ENG DUAL FAILURE - FUEL REMAINING	A
Engine relight in flight, windmill relight	A

Emergency electrical configuration

A B

Conditional

A performs this in either branch; if both trainees are co-pilots, remove the malfunction after procedures. B performs it only if both trainees are captains.

UNRELIABLE SPEED INDICATION, demo of BUSS & BAS

B

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTSG / VTSG
ALTERNATE	VTSP
FLIGHT NUMBER	THA05
COST INDEX	30
PLANNED CRUISE	FL160
FLAPS	1 or 2
ZFW (T)	55
ZFWCG (%MAC)	26.1
TAKE-OFF FUEL (T)	10
TOW (T)	65
TRIP FUEL (T)	—
TAKE-OFF CG (%MAC)	26.1
ROUTE	—

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTSG
RUNWAY	32
WIND	230/20
VISIBILITY	3 km
CEILING	1500
TEMP °C	30
QNH	1010
RWY COND.	Dry

FIELD	BOTH TRAINEES
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTSG
RUNWAY	32
WIND	050/20
VISIBILITY	3 km
CEILING	1500
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Take-off position	
2	Take-off - engine failure after reaching V2, familiarization	A only
3	Take-off - engine failure between V1 and V2, repeat to proficiency	Both
4	Take-off - engine failure after reaching V2	Both
5	Engine relight in flight; reposition FL350	Both
6	ENG 1(2) FAIL, drift down procedure	A only
7	ENG DUAL FAILURE - FUEL REMAINING	A only
8	Engine relight in flight, windmill relight	A only
9	Radar vectoring for RNP approach	A only
10	Approach - 3D RNP manually with FD, degradation or system failure	A only
11	Go-around at minima, engine failure at go-around	A only
12	Follow ATC instructions for ILS approach	A only
13	Approach - ILS manually with FD, one engine inoperative	Both
14	Landing - one engine inoperative; repeat single-engine landing	Both
15	Take-off - crosswind	Both

#	TOPIC	A/B
16	ELEC EMER CONFIG	A only
17	Radar vectoring for ILS approach	
18	Approach - ILS raw data, manually without FD	Both
19	Landing - crosswind	Both

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off position	
2	Take-off – engine failure between V1 and V2, repeat to proficiency	Both
3	Take-off – engine failure after reaching V2	Both
4	Engine relight in flight	Both
5	Radar vectoring, for ILS approach	
6	Approach – ILS, manually, with FD	B only
7	Go-around – during instrument approach	B only
8	ENG 1(2) FAIL, during go-around	B only
9	Follow ATC instructions, for the same ILS as above	B only
10	Approach – ILS, manually, with FD, one engine inoperative	Both
11	Landing – one engine inoperative; reposition to final for single-engine landing practice	Both
12	Take-off – crosswind	Both
13	UNRELIABLE SPEED INDICATION, demo of BUSS & BAS	B only
14	Radar vectoring, for ILS approach	
15	Approach – ILS raw data, manually, without FD	Both
16	Landing – crosswind	Both

Conditional alternate

TRAINEE A · BOTH CO-PILOTS

1. Both trainees co-pilots: ELEC EMER CONFIG; remove after procedures complete
2. Radar vectoring for ILS approach
3. Approach - ILS raw data, manually without FD
4. Landing - crosswind

TRAINEE B · BOTH CAPTAINS

1. Both trainees captains: ELEC EMER CONFIG
2. Radar vectoring for ILS approach
3. Approach - ILS raw data, manually without FD
4. Landing - crosswind

SESSION 14 OF 18

FFS 6

A VTSP / VTSP | B VTSP / VTSP

Combined practice 19 items

TAKEOFF & DEPARTURE

Take-off - max crosswind	Both
Windshear - take-off	Both
Take-off - crosswind	Both A ×2
Rejected take-off - engine fire below V1	Both

APPROACHES & GO-AROUNDS

Approach - ILS manually with FD	Both
Approach - 3D RNP, with AP, LNAV/VNAV minima, FINAL APP	A

LANDING

Landing with partial/no flaps or slats A repeats this landing across hydraulic and flap-fault scenarios.	Both A ×3
Landing - jammed horizontal stabilizer, out of trim	Both

FAILURES & SYSTEMS

Follow SID, RNAV, RNP degradation or failure	Both
ANTI-ICE CAPT PROBES; insert TCAS as failure appears	Both
TCAS RA event	Both
HYD G+B SYS LO PR	A
HYD G+Y SYS LO PR, handling	Both
F/CTL FLAPS FAULT/LOCKED	A
Stabilizer jam	A
Emergency evacuation	Both
L/G gravity extension	B

F/CTL SLATS FAULT/LOCKED, during retraction B

FLIGHT HANDLING

Visual circuits Both

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTSP / VTSP
ALTERNATE	VTSG
FLIGHT NUMBER	THA06
COST INDEX	30
PLANNED CRUISE	FL160
FLAPS	1 or 2
ZFW (T)	46
ZFWCG (%MAC)	35
TAKE-OFF FUEL (T)	10
TOW (T)	56
TRIP FUEL (T)	—
TAKE-OFF CG (%MAC)	32.4
ROUTE	—

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTSP
RUNWAY	27
WIND	200/15
VISIBILITY	3 km LT RA
CEILING	1500
TEMP °C	25
QNH	1004
RWY COND.	Wet

FIELD	BOTH TRAINEES
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTSP
RUNWAY	27
WIND	200/15
VISIBILITY	3 km LT RA
CEILING	1500
TEMP °C	25
QNH	1004
RWY COND.	Wet
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Take-off position	
2	Take-off - max crosswind	Both
3	Windshear - take-off	Both
4	Follow SID, RNAV, RNP degradation or failure	Both
5	ANTI-ICE CAPT PROBES; insert TCAS as failure appears	Both
6	TCAS RA event	Both
7	HYD G+B SYS LO PR	A only
8	Follow ATC instructions for ILS approach	
9	Approach - ILS manually with FD	Both
10	Landing - partial/no flaps/slats; reposition and set G+Y hydraulic fault	Both
11	HYD G+Y SYS LO PR, handling	Both
12	Landing - partial/no flaps/slats	Both
13	Take-off - crosswind	Both
14	Follow ATC instructions for RNP approach	A only
15	F/CTL FLAPS FAULT/LOCKED	A only

#	TOPIC	A/B
16	Approach - 3D RNP, with AP, LNAV/VNAV minima, FINAL APP	A only
17	Landing - partial/no flaps/slats	Both
18	Take-off - crosswind	Both
19	Stabilizer jam	A only
20	Visual circuits	Both
21	Landing - jammed horizontal stabilizer, out of trim	Both
22	Rejected take-off - engine fire below V1	Both
23	Emergency evacuation	Both

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off position	
2	Take-off – max crosswind	Both
3	Windshear – take-off	Both
4	Follow SID, RNAV, climb SID altitude/FL, RNP degradation or system failure	Both
5	ANTI-ICE CAPT PROBES; insert a TCAS as failure appears	Both
6	TCAS RA event	Both
7	HYD G + Y SYS LO PR	Both
8	Follow ATC instructions, for ILS approach	
9	L/G gravity extension	B only
10	Approach – 3D (ILS), manually, with FD	Both
11	Landing – jammed horizontal stabilizer in any out-of-trim position	Both
12	Take-off – crosswind	Both
13	F/CTL SLATS FAULT/LOCKED, during retraction	B only
14	Radar vectoring, for a visual approach	B only
15	Visual circuits	Both
16	Landing – without extended or with partly extended flaps/slats	Both
17	Rejected take-off – engine fire below V1	Both
18	Emergency evacuation	Both

SESSION 15 OF 18

FFS 7

A VTBS / VHHH | B VHHH / VTBS

Combined practice 27 items

TAKEOFF & DEPARTURE

Rejected take-off due to predictive windshear	Both
Flight crew incapacitation during rejected take-off	Both
Takeoff at maximum mass in crosswind A also includes an ARS demonstration at 210 kt.	Both
Take-off - engine severe damage/separation between V1 and V2, night	A
Windshear – take-off	B
Take-off – engine failure between V1 and V2, during night conditions	B

APPROACHES & GO-AROUNDS

Approach - ILS raw data to 200 ft DH/A, overweight	A
Go-around at DH/A, overweight	A
Approach - ILS manually without FD	Both
3D RNP approach to LNAV/VNAV minima with degraded RNP A specifies that RNP degradation leads to a go-around; B lists degradation or system malfunction.	Both
Discontinued approach	A
ILS approach manually with FD, one engine inoperative A specifies the engine inoperative before passing 1,000 ft AAL.	Both
Go-around – at minima, one engine inoperative; follow missed approach procedure	B

LANDING

OVERWEIGHT LANDING B combines the overweight landing with manual ILS.	Both
Landing - one engine inoperative, crosswind	Both
LOSS OF BRAKING	Both

FAILURES & SYSTEMS

AIR PACK 1(2) OFF; select ON per SOP	Both
AUTO FLT FCU 1+2 FAULT	Both
Fuel imbalance	A
APU AUTO (EMER) SHUT DOWN	Both
APU fire B combines this with APU emergency shutdown.	B
Fuel leak	B
ELEC APU GEN FAULT	B
Removal of smoke/fumes, due to smoke in AFT cabin	B
Emergency evacuation	B
Engine relight in flight	B

FLIGHT HANDLING

Holding procedures	Both

Flight setup

FIELD	TRAINEE A	TRAINEE B
FROM / TO DIFFERENT	VTBS / VHHH	VHHH / VTBS
ALTERNATE DIFFERENT	VMMC	VTBD
FLIGHT NUMBER	THA07	THA07
COST INDEX	30	30
PLANNED CRUISE DIFFERENT	FL330	FL340
FLAPS	1 or 2	1 or 2
ZFW (T)	56	56
ZFWCG (%MAC)	30.5	30.5
TAKE-OFF FUEL (T)	13	13
TOW (T)	69	69
TRIP FUEL (T) DIFFERENT	—	7.5
TAKE-OFF CG (%MAC)	28.7	28.7

FIELD	TRAINEE A	TRAINEE B
ROUTE DIFFERENT	VTBSR19 SELKA1J SELKA A1 TOPER W42 PAKRI DCT GRASO Y15 GUROK A202 SIKOU V571 CANTO CANTO3A VHHHR07C	VHHHR25L PECAN2B PECAN V10 SIKOU A202 SAV B460 ROT R345 GRASO Y20 KRT DCT RUKSA EASTE1C VTBSR19

Flight conditions

DEPARTURE

FIELD	A	B
AIRPORT DIFFERENT	VTBS	VHHH
RUNWAY DIFFERENT	19	25L
WIND DIFFERENT	110/20	170/20
VISIBILITY	3 km	3 km
CEILING	1000	1000
TEMP °C	30	30
QNH	1010	1010
RWY COND.	Wet	Wet
ANTI-ICE	Off	Off

ARRIVAL

FIELD	A	B
AIRPORT DIFFERENT	VHHH	VTBS
RUNWAY DIFFERENT	07C	19
WIND DIFFERENT	010/20	110/20
VISIBILITY	3 km	3 km
CEILING	1000	1000
TEMP °C	30	30
QNH	1020	1020
RWY COND.	Wet	Wet
ANTI-ICE	Off	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Take-off position	A only

#	TOPIC	A/B
2	Rejected take-off due to predictive windshear	Both
3	Flight crew incapacitation during rejected take-off	Both
4	Take-off - max mass, crosswind; ARS demo at 210 kt	Both
5	AIR PACK 1(2) OFF; select ON per SOP	Both
6	Follow ATC instructions for ILS approach	A only
7	AUTO FLT FCU 1+2 FAULT	Both
8	Approach - ILS raw data to 200 ft DH/A, overweight	A only
9	Go-around at DH/A, overweight	A only
10	Approach - ILS manually without FD	Both
11	OVERWEIGHT LANDING	Both
12	Take-off - engine severe damage/separation between V1 and V2, night	A only
13	Holding procedures	Both
14	Fuel imbalance	A only
15	Approach - 3D RNP, with AP, LNAV/VNAV minima; degradation leads to go-around	Both
16	Discontinued approach	A only
17	Radar vectoring for ILS or RNP approach	A only
18	APU AUTO (EMER) SHUT DOWN	Both
19	Approach - ILS manually with FD, one engine inoperative before 1000 ft AAL	Both
20	Landing - one engine inoperative, crosswind	Both
21	LOSS OF BRAKING	Both

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off	B only
2	Rejected take-off, due to predictive windshear	Both
3	Incapacitation of a flight crew member, during rejected take-off	Both
4	Take-off – max TO mass, crosswind	Both
5	Windshear – take-off	B only
6	AIR PACK 1(2) OFF, to be selected ON as per SOP	Both
7	Follow ATC instructions, for RNP approach	B only
8	Fuel leak	B only

#	TOPIC	A/B
9	Holding procedures	Both
10	Approach – 3D (RNP), with AP, to LNAV/VNAV minima, RNP degradation or system malfunction	Both
11	Go-around – at minima, one engine inoperative; follow missed approach procedure	B only
12	Radar vectoring, for ILS approach	B only
13	ELEC APU GEN FAULT	B only
14	APU AUTO (EMER) SHUT DOWN, APU FIRE	B adds
15	Removal of smoke/fumes, due to smoke in AFT cabin	B only
16	Approach – ILS, manually, with FD, one engine inoperative	Both
17	Landing – one engine inoperative, crosswind	Both
18	Emergency evacuation	B only
19	Take-off – engine failure between V1 and V2, during night conditions	B only
20	Engine relight in flight	B only
21	AUTO FLT FCU 1+2 FAULT	Both
22	Approach – ILS, manually, without FD, overweight; OVERWEIGHT LANDING	Both
23	LOSS OF BRAKING	Both

SESSION 16 OF 18

FFS 8

A VTBS / VTBS | B VTBS / VTBS

Combined practice 20 items

TAKEOFF & DEPARTURE

Normal take-off, different flap settings; expedited climb to 5000 ft	Both

FLIGHT HANDLING

Roll rate with full aileron/spoiler input	Both
Roll rate with rudder input	Both
Pitch change using only stabilizer trim	Both
Pitch change using thrust adjustments	Both
Pitch change using speed brakes	Both
Yaw/roll from asymmetric thrust, with autopilot	Both
Yaw/roll from asymmetric thrust, without autopilot	Both
Approach-to-stall recovery using only pitch control	Both
Fly-by-wire qualities	Both
Exploring jet engine performance	Both
Exercise 1 nose high: nose-down elevator	Both
Exercise 2 nose high: use of bank angle	Both
Exercise 3 nose high: thrust reduction	Both
Exercise 4 nose low: nose-low recovery	Both
Exercise 5 nose low: accelerated stall demonstration	Both
Exercise 6 nose low: high bank/inverted flight	Both
Exercise 7 nose high, low energy	Both
Exercise 8 high-altitude stall warning	Both
Exercise 9 landing configuration stall	Both

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTBS
ALTERNATE	VTBD
FLIGHT NUMBER	THA08
COST INDEX	30
PLANNED CRUISE	—
FLAPS	1 or 2
ZFW (T)	52
ZFWCG (%MAC)	28.6
TAKE-OFF FUEL (T)	9
TOW (T)	61
TRIP FUEL (T)	—
TAKE-OFF CG (%MAC)	27.3
ROUTE	—

Flight conditions

DEPARTURE

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19
WIND	Calm
VISIBILITY	CAVOK
CEILING	—
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

ARRIVAL

FIELD	BOTH TRAINEES
AIRPORT	VTBS
RUNWAY	19

FIELD	BOTH TRAINEES
WIND	Calm
VISIBILITY	CAVOK
CEILING	—
TEMP °C	30
QNH	1010
RWY COND.	Dry
ANTI-ICE	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Take-off	
2	Normal take-off, different flap settings; expedited climb to 5000 ft	Both
3	Roll rate with full aileron/spoiler input	Both
4	Roll rate with rudder input	Both
5	Pitch change using only stabilizer trim	Both
6	Pitch change using thrust adjustments	Both
7	Pitch change using speed brakes	Both
8	Yaw/roll from asymmetric thrust, with autopilot	Both
9	Yaw/roll from asymmetric thrust, without autopilot	Both
10	Approach-to-stall recovery using only pitch control	Both
11	Fly-by-wire qualities	Both
12	Exploring jet engine performance	Both
13	Exercise 1 nose high: nose-down elevator	Both
14	Exercise 2 nose high: use of bank angle	Both
15	Exercise 3 nose high: thrust reduction	Both
16	Exercise 4 nose low: nose-low recovery	Both
17	Exercise 5 nose low: accelerated stall demonstration	Both
18	Exercise 6 nose low: high bank/inverted flight	Both
19	Exercise 7 nose high, low energy	Both
20	Exercise 8 high-altitude stall warning	Both

#	TOPIC	A/B
21	Exercise 9 landing configuration stall	**Both**

TRAINEE B

#	TOPIC	A/B
1	INIT: Take-off	
2	Normal take-off, different flap settings, incl. expedited climb to 5000 ft	**Both**
3	Roll rate with full aileron/spoiler input	**Both**
4	Roll rate with rudder input	**Both**
5	Pitch change using only stabilizer trim	**Both**
6	Pitch change using thrust adjustments	**Both**
7	Pitch change using speed brakes	**Both**
8	Yaw motion and resultant roll due to asymmetric thrust, with autopilot	**Both**
9	Yaw motion and resultant roll due to asymmetric thrust, without autopilot	**Both**
10	Approach-to-stall recovery (stick shaker) using only pitch control	**Both**
11	Fly-by-wire qualities	**Both**
12	Exploring jet engine performance	**Both**
13	Exercise 1: Nose high · Iteration 1: use of nose-down elevator	**Both**
14	Exercise 2: Nose high · Iteration 2: use of bank angle	**Both**
15	Exercise 3: Nose high · Iteration 3: thrust reduction (underwing-mounted engines)	**Both**
16	Exercise 4: Nose low · Iteration 4: nose-low recovery	**Both**
17	Exercise 5: Nose low · Iteration 5: accelerated stall demonstration	**Both**
18	Exercise 6: Nose low · Iteration 6: high bank / inverted flight	**Both**
19	Exercise 7: Nose-high, low energy	**Both**
20	Exercise 8: High-altitude stall warning	**Both**
21	Exercise 9: Landing configuration stall	**Both**

SESSION 17 OF 18

FFS 9

A VTBS / VCBI | B VCBI / VTBS

Combined practice 15 items

TAKEOFF & DEPARTURE

Take-off - normal	Both

APPROACHES & GO-AROUNDS

Approach - 3D (ILS), with AP	Both
Go-around - 3D at DH/A; follow missed approach	Both
Approach - 2D LOC or LNAV, down to MDH/A	Both

LANDING

Landing - crosswind	Both

FAILURES & SYSTEMS

TCAS RA event	Both
At 15 min to EEP: IDG 2 OVHT; use APU	A
Entering ETOPS area: navigation failure, both FMGS fault	A
At ETP: engine 1 or 2 fail; strategies	A
Diversion	Both
Fuel monitoring and imbalance	Both
At 15 min to EEP – HYD G RSVR LO LVL – restore	B
Entering ETOPS area – ALL ADR FAULT – restore	B
At ETP – LEFT WING DUCT LEAK – PACK 2 OVHT	B
Cargo smoke	B

Flight setup

FIELD	TRAINEE A	TRAINEE B
FROM / TO DIFFERENT	VTBS / VCBI	VCBI / VTBS
ALTERNATE DIFFERENT	VOMM	VTBD
FLIGHT NUMBER	THA 9	THA 9
COST INDEX	30	30
PLANNED CRUISE	Unspecified	Unspecified
FLAPS	1 or 2	1 or 2
ZFW (T)	60	60
ZFWCG (%MAC)	29.5	29.5
TAKE-OFF FUEL (T)	12	12
TOW (T)	72	72
TRIP FUEL (T)	Unspecified	Unspecified
TAKE-OFF CG (%MAC)	28.1	28.1
ROUTE	Unspecified	Unspecified

Flight conditions

DEPARTURE

FIELD	A	B
AIRPORT DIFFERENT	VTBS	VCBI
RUNWAY DIFFERENT	19	04
WIND	110/15	110/15
VISIBILITY DIFFERENT	5 km	5 km LT RA
CEILING DIFFERENT	2000	1500
TEMP °C DIFFERENT	28	25
QNH DIFFERENT	1008	1005
RWY COND. DIFFERENT	Dry	Wet
ANTI-ICE	Off	Off

ARRIVAL

FIELD	A	B
AIRPORT DIFFERENT	VCBI	VTBS
RUNWAY DIFFERENT	04	19

FIELD	A	B
WIND	110/15	110/15
VISIBILITY DIFFERENT	5 km LT RA	5 km
CEILING DIFFERENT	1500	2000
TEMP °C DIFFERENT	25	28
QNH DIFFERENT	1005	1008
RWY COND. DIFFERENT	Wet	Dry
ANTI-ICE	Off	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	Preflight procedures	A only
2	Engine start	
3	Taxiing	
4	Before take-off checks	
5	Take-off - normal	Both
6	Follow SID	
7	TCAS RA event	Both
8	Climb to cleared FL	
9	Cruise	
10	At 15 min to EEP: IDG 2 OVHT; use APU	A only
11	Entering ETOPS area: navigation failure, both FMGS fault	A only
12	At ETP: engine 1 or 2 fail; strategies	A only
13	Diversion	Both
14	Fuel monitoring and imbalance	Both
15	Follow STAR or radar vectoring	
16	Approach - 3D (ILS), with AP	Both
17	Go-around - 3D at DH/A; follow missed approach	Both
18	Radar vectoring	
19	Approach - 2D LOC or LNAV, down to MDH/A	Both
20	Landing - crosswind	Both

#	TOPIC	A/B
21	After landing procedures - shutdown	

TRAINEE B

#	TOPIC	A/B
1	Preflight preparation	**B only**
2	Engine start	
3	Taxiing	
4	Before take-off checks	
5	Take-off – normal	**Both**
6	Follow SID	
7	TCAS RA event	**Both**
8	Climb to cleared FL	
9	Cruise	
10	At 15 min to EEP – HYD G RSVR LO LVL – restore	**B only**
11	Entering ETOPS area – ALL ADR FAULT – restore	**B only**
12	At ETP – LEFT WING DUCT LEAK – PACK 2 OVHT	**B only**
13	Cargo smoke	**B only**
14	Diversion	**Both**
15	Fuel monitoring and imbalance	**Both**
16	Follow STAR or radar vectoring	
17	Approach – 3D (ILS), with AP	**Both**
18	Go-around – 3D, at DH/A, follow missed approach procedure	**Both**
19	Radar vectoring	
20	Approach – 2D, down to MDH/A, LOC or LNAV	**Both**
21	Landing – crosswind	**Both**
22	After landing procedures – shutdown	

SESSION 18 OF 18

FFS 10

A VTBS / VTCC | B VTBS / VTCC

Combined practice 19 items

TAKEOFF & DEPARTURE

Engine start fault	Both
Take-off - SID	Both
Take-off - windshear	Both
Take-off with one engine inoperative between V1 and VR	Both
Take-off - engine fire before V1	Both
Rejected take-off before V1 B specifies a reasonable speed below V1.	Both

APPROACHES & GO-AROUNDS

2D approach to MDA (VTCC)	Both
Radar vectors - 3D (ILS) approach to DA	Both
One engine inoperative manual go-around	Both
One engine inoperative visual approach	Both

LANDING

Landing - crosswind	Both
Landing without extended flaps	Both
One engine inoperative landing	Both

FAILURES & SYSTEMS

Climb - TCAS event	Both
Cruise - decompression / emergency descent	Both
Hydraulic systems G+Y failed (low level)	Both
Flaps system fault	Both

Manual flying without FD	Both
Emergency evacuation	Both

Flight setup

FIELD	BOTH TRAINEES
FROM / TO	VTBS / VTCC
ALTERNATE	VTCT
FLIGHT NUMBER	THA10
COST INDEX	30
PLANNED CRUISE	FL330
FLAPS	1 or 2
ZFW (T)	49.5
ZFWCG (%MAC)	30.5
TAKE-OFF FUEL (T)	8
TOW (T)	57.5
TRIP FUEL (T)	2.8
TAKE-OFF CG (%MAC)	32.4
ROUTE	VTBSR19 OLVUK1J OLVUK Y26 MARNI MARNI2A VTCCR36

Flight conditions

DEPARTURE

FIELD	A	B
AIRPORT	VTBS	VTBS
RUNWAY	19	19
WIND	110/20	110/20
VISIBILITY DIFFERENT	3 km LT RA	5 km
CEILING	1500	1500
TEMP °C DIFFERENT	28	30
QNH DIFFERENT	1005	1008
RWY COND. DIFFERENT	Wet	Dry
ANTI-ICE	Off	Off

ARRIVAL

FIELD	A	B
AIRPORT	VTCC	VTCC
RUNWAY	36	36
WIND	090/20	090/20
VISIBILITY	3 km LT RA	3 km LT RA
CEILING	2500	2500
TEMP °C DIFFERENT	25	28
QNH DIFFERENT	1004	1010
RWY COND.	Wet	Wet
ANTI-ICE	Off	Off

Ordered session topics

TRAINEE A

#	TOPIC	A/B
1	INIT: Gate position (VTBS)	
2	Cockpit preparation, before start, engine start	
3	Engine start fault	Both
4	Taxi, before take-off	
5	Take-off - SID	Both
6	Climb - TCAS event	Both
7	Cruise - decompression / emergency descent	Both
8	2D approach to MDA (VTCC)	Both
9	Landing - crosswind	Both
10	INIT: Take-off position (VTBS)	
11	Take-off - windshear	Both
12	Hydraulic systems G+Y failed (low level)	Both
13	Flaps system fault	Both
14	Manual flying without FD	Both
15	Landing without extended flaps	Both
16	INIT: Take-off position (VTBS)	
17	Take-off with one engine inoperative between V1 and VR	Both

#	TOPIC	A/B
18	Radar vectors - 3D (ILS) approach to DA	Both
19	One engine inoperative manual go-around	Both
20	One engine inoperative visual approach	Both
21	One engine inoperative landing	Both
22	INIT: Take-off position (VTBS)	
23	Take-off - engine fire before V1	Both
24	Rejected take-off before V1	Both
25	Emergency evacuation	Both

TRAINEE B

#	TOPIC	A/B
1	INIT: Gate position (VTBS)	
2	Cockpit preparation, before start, ENG start	
3	Engine start fault	Both
4	Taxi, before take-off	
5	Take-off – SID	Both
6	Climb – TCAS event	Both
7	Cruise – decompression – emergency descent	Both
8	2D approach to MDA (VTCC)	Both
9	Landing – crosswind	Both
10	INIT: Take-off position (VTBS)	
11	Take-off – windshear	Both
12	Hydraulic systems G+Y failed (low level)	Both
13	Flaps system fault	Both
14	Manual flying without FD	Both
15	Landing without extended flaps	Both
16	INIT: Take-off position (VTBS)	
17	Take-off and one engine inoperative between V1 and VR	Both
18	Radar vectors – 3D (ILS) approach to DA	Both
19	One engine inoperative manual go-around	Both
20	One engine inoperative visual approach	Both

#	TOPIC	A/B
21	One engine inoperative landing	**Both**
22	INIT: Take-off position (VTBS)	
23	Take-off – engine fire (before V1)	**Both**
24	Rejected take-off at a reasonable speed before V1	**Both**
25	Emergency evacuation	**Both**